Hybrid Zone: Bridging Acoustic and Wi-Fi for Enhanced Gesture Recognition

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***Abstract*—Gesture recognition has significant applications such as assisted living, e-health, and human-device interactions. Moving away from conventional computer vision techniques, recent studies are turning towards ubiquitous methods such as Wi-Fi and acoustic signals, which offer device deployment at minimal costs. In this paper, we explore these two ubiquitous techniques to enhance gesture recognition, addressing challenges related to multi-modal fusion. Due to the inherent differences in signal types, we employ a tailored method to harmonize information from these distinct signals. Traditional multi-modal fusion methods often lack theoretical models due to insufficient analysis of the foundational characteristics of different signals. In particular, we propose a concept, Hybrid Zone, which is a theoretical model illustrating the fusion of acoustic and Wi-Fi sensing. Hybrid zone offers a comprehensive perspective on the fusion of acoustic and Wi-Fi sensing areas. Moreover, it provides an intricate view of the synthesis of acoustic and Wi-Fi velocities at a granular level. Our experimental results have been promising, achieving a high accuracy rate of 93.75% in gesture recognition.**

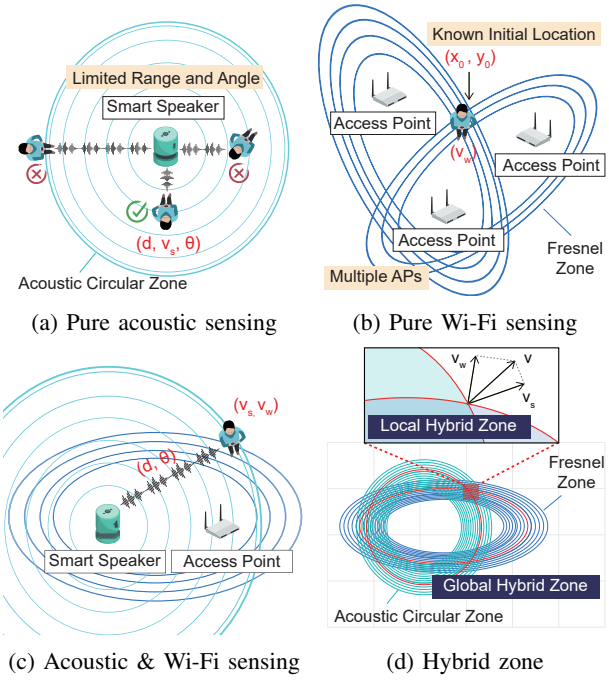
***Index Terms*—Acoustic, Wi-Fi, Gesture recognition**

I. Introduction

A. Background and Motivation

The gesture recognition market, without requiring physical touch, is projected to reach 37.6 billion by 2026 [1]. The growth in non-contact sensing technology has enabled smart devices to detect human actions without direct contact, enhancing human-computer interaction by providing more natural and intuitive control methods. In this context, Wi-Fi [2]–[6] and acoustic sensing [7]–[9], both integral components of household wireless technologies, are receiving increased focus. Their ubiquity in homes, serving as the backbone for internet and audio systems, renders them highly suitable for everyday applications in non-contact human action detection. Despite their potentials, standalone sensing methods have limitations: i) Acoustic systems [10] are advantageous in low-light or obstructed environments, yet they are susceptible to ambient noise and echoes [7], [8], and their accuracy decreases with distance. These systems also necessitate a direct line-of-sight to the speaker. ii) Wi-Fi based systems offer longer sensing ranges compared to acoustic systems. However, they require specific knowledge of the user's location for accurate feature extraction, and a minimum of three Wi-Fi access points are needed for location-independent recognition [2], [11], [12].

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(a) Pure acoustic sensing (b) Pure Wi-Fi sensing

(c) Acoustic & Wi-Fi sensing (d) Hybrid zone

Fig. 1: Rapid movement affects distance assessment accuracy.

These challenges in pure technology-based gesture recognition led us to investigate the fusion of acoustic and Wi-Fi sensing technologies. This approach can be practically implemented using common household devices such as smart speakers and Wi-Fi routers, which are typically interconnected. This fusion aims to overcome the individual limitations of each technology, offering a more robust and accurate gesture recognition system. Naturally, we explore the fusion of acoustic and Wi-Fi sensing technologies in Fig. 1(c). It is practical to build the described fusion sensing scenarios using just one pair of devices, such as the smart speaker and the router, because voice devices commonly found in our homes can usually connect to the Wi-Fi router.

Currently, many cross-modal fusion approaches rely heavily on the cross-attention mechanism [13]–[15], which can yield excellent results. However, they suffer from three main issues: i) the need for manual collection of training datasets, ii) lack of explanation for the underlying mechanisms, and iii) difficulty in ensuring the generalization capability of the system. Some recent works have explored the fusion of Wi-Fi and acoustics

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in the model [16], but they require targets or users to actively emit sounds (*e.g.*, footsteps). However, when making gestures, people do not produce sounds, so acoustic devices are needed to actively emit inaudible signals to perceive the surrounding targets. In contrast to sound source localization methods [16], our contact-free sensing approach focuses on analyzing signals reflected from targets. Nevertheless, compared to direct signals, reflected signals have weaker strength, presenting a greater challenge in achieving this objective.

B. Challenges and Solutions

This paper presents *Hybrid Zone*, a theoretical model to illustrate how to fuse acoustic and Wi-Fi sensing techniques. The basic principle consists of global and local stages: i) The global hybrid zone defines the overall distribution of the hybrid sensing signal. We leverage the acoustic signal to approximate the target's position, enabling us to focus on the specific area of interest, referred to as the local Hybrid Zone; ii) At the local stage, the Hybrid Zone adopts a detailed approach to accurately synthesize user velocity. This is achieved by combining the velocity measurements from both acoustic and Wi-Fi signals, leading to enhanced precision in tracking and recognition. The study addresses two main challenges in the implementation of this model.

Challenge 1: How to obtain accurate acoustic features in non-ideal situations? Specifically, non-ideal situations include long distances, fast motion, and indirect orientation. In the case of long-distance and non-direct directions, the resulting signal is weaker, making it harder to accurately extract features. Additionally, measuring the distance of a fast-moving target can be challenging due to the Doppler effect, which causes an extra frequency shift in the reflected signal that hinders the measurement of the distance. To address these problems, we first utilize the triangular chirp signal to distinguish the Doppler effect and the target's distance. Next, we design a signal enhancement method so that the weak reflected signal can be enhanced for accurate feature extraction.

Challenge 2: How to design a theoretical model to fuse multimodal information for fine-grained behavior tracking? Previous research has typically employed neural networks, such as the cross-attention mechanism [14], to combine sensing features from different modalities. However, there is a gap in the theoretical understanding of how these modalities are integrated. In order to provide theoretical guidance for the fusion of Wi-Fi and acoustic sensing technology, we propose the concept of hybrid zone. The global hybrid zone has revealed that the sensing zone is composed of a cluster of ellipses and a cluster of circles. The altered features with respect to the object are caused by its intersection with this zone. The local hybrid zone reveals that the detected velocity features are the components of the actual velocity on the circle and ellipse normal vectors, thus providing a theoretical velocity synthesis model.

C. Contributions and Advantages

In this paper, we make the following contributions:

- To the best of our knowledge, hybrid zone for the first time provides a theoretical explanation for the fusion of acoustic and Wi-Fi sensing zones. Through the design of both global and local hybrid zones, we can fully leverage the benefits of acoustic and Wi-Fi sensing to achieve precise gesture recognition in a complementary way.
- In order to obtain stable acoustic features, we propose a series of methods at the signal processing level, including the triangular chirp analysis and signal enhancement, so that accurate position-related features can be resolved even in non-ideal situations.
- We implement the prototype with commercial off-the-shelf (COTS) Wi-Fi and acoustic devices. The experimental results demonstrate that we can realize 93.75% accuracy of gesture recognition.

With the proposed technique contribution, our system offers a distinct advantage over previous systems: voice-controlled devices typically have the ability to connect to the Wi-Fi router, which allows for the complementary use of Wi-Fi and acoustics without extra constraints. Furthermore, constructing a theoretical model can enhance our understanding of the sensing mechanism and provide theoretical guidance for the practical implementation of the system.

II. Related Works

In this study, our methodology avoids many of the pitfalls typically encountered in single-data-source systems. As displayed in TABLE I, we have selected specific criteria for assessing the efficacy of these methods. These criteria encompass their capability to identify swift motions, offer theoretical analysis, and extract location-independent attributes.

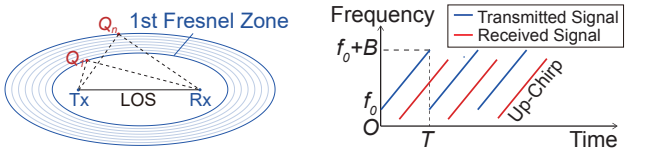
Wi-Fi Sensing System. Wi-Fi-based gesture recognition systems operate by identifying human gesture-induced alterations in Wi-Fi signal paths. Each gesture generates distinctive CSI variations, which are learned and recognized by machine learning algorithms. However, their precision can be influenced by device placement, movements of other objects, and shifts in the environment. In particular, Widar3.0 [12] establishes a theoretical model and proposes a body velocity profile (BVP) for cross-domain human gesture recognition. However, it requires multiple Wi-Fi devices (*e.g.*, 6 APs) and a known user's location to extract location-independent features. OneFi [17] designs a lightweight few-shot learning framework to recognize unseen gestures with a shorter training time. WiGr [18] uses a domain-transferable mapping mechanism and a dual-path prototypical network to extract domain-independent features for cross-domain gesture recognition. However, the reliance on neural networks [2], [17]–[20] in these methods is excessive and they lack an analysis of the internal mechanism of sensing ability, making it challenging to determine their scalability.

Acoustic Sensing System. Acoustic-based gesture recognition systems leverage sound waves, frequently in the ultrasonic spectrum, to discern gestures. These systems employ a speaker to emit sound waves, with the microphone array detecting any gesture-induced alterations. Subsequently, machine learning

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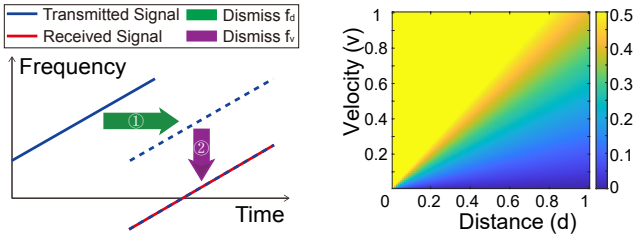
TABLE I: Comparison of different gesture recognition systems.

System	Data Source	Room level?	Fast Move?	Theoretical Model?	Location-Independent?	Single Link?
Widar3.0 2021 [12]	Wi-Fi	Yes	Yes	Yes	Yes	No
OneFi 2021 [17]	Wi-Fi	Yes	No	No	Yes	Yes
WiGr 2022 [18]	Wi-Fi	Yes	No	No	Yes	Yes
FM-Track 2022 [7]	Acoustic	No	No	Yes	Yes	Yes
LASense 2022 [8]	Acoustic	Yes	No	Yes	Yes	Yes
RTrack 2022 [9]	Acoustic	Yes	No	Yes	Yes	Yes
Our Approach	Wi-Fi + Acoustic	Yes	Yes	Yes	Yes	Yes



(a) The schematic diagram of Fresnel zone. (b) Transmitted and received signals in linear up-chirps.

Fig. 2: Theoretical models in Wi-Fi and acoustic sensing.



(a) Shifting for dismissing the distance (f_d) and velocity (f_v). (b) Proportion of velocity's influence on shift relative to distance.

Fig. 3: Rapid movement affects distance assessment accuracy.

algorithms correlate these wave patterns to particular gestures. Notably, these systems operate independently of visual input, proving beneficial in low-light or visually obstructed scenarios, though their efficiency can be influenced by environmental noise or echo. MilliSonic [21] enhances acoustic-based tracking with a unique algorithm that ensures sub-millimeter precision. Mao *et al.* [10] proposed a joint estimation of the signal's propagation distance and angle of arrival (AoA), derived from the 2D MUSIC algorithm. Symphony [22] uses microphone geometry and signal path characteristics to efficiently identify and distinguish signal sources. MAVL [23] is a human voice localization system, which first estimates the room structure and then improves a 3D MUSIC algorithm. Li *et al.* [9] expanded the range of acoustic sensing, demonstrating the possibility of room-scale hand gesture recognition using standard smart speakers. While acoustic methods effectively recognize certain movements, particularly in the tangential direction to the sensor, they struggle with complex, rapid, or multi-dimensional movements. This limitation underscores the need for integrating other sensing technologies for more robust and accurate tracking of fast, remote movements.

III. Preliminaries and Insights

A. Fundamentals

The purpose of this paper is to propose a theoretical model that combines the use of both Wi-Fi and acoustic sensing. To this end, we will respectively introduce the theoretical models of these two techniques.

1) The theoretical model for Wi-Fi sensing : The Fresnel zone model [24]–[26] is a crucial foundational framework in Wi-Fi sensing, as it helps to explain the mechanisms behind human activity sensing through Wi-Fi signals. To be more specific, the impact of human behavior on Wi-Fi signals is reflected in the change in the length of the signal path reflected by the human body, which is $|T_xQ_n|+|Q_nR_x|$ in Fig. 2(a). We will observe a complete periodic variation in the magnitude of Channel State Information (CSI) as the length of the reflected path varies by one wavelength λ. The Fresnel zone reveals that points whose reflection paths are constant will form the ellipse satisfying $|T_xQ_n| + |Q_nR_x| - |T_xR_x| = n\frac{\lambda}{2}$, where Q_n is a point on the n-th ellipsoid. Moreover, the length of the reflected path between adjacent ellipses changes by half a wavelength, which corresponds to a half-period change in the magnitude of CSI.

2) The theoretical model for acoustic sensing: The fundamental principle of acoustic sensing is to actively emit inaudible Frequency-Modulated Continuous Wave (FMCW) signals and then analyze how the behavior of a person changes the features of the FMCW signal [7]. Generally, these systems [8] typically use chirp signals in which the signal's frequency changes linearly as illustrated in Fig. 2(b). When the chirp signals are reflected by the object, the time delay between the transmitted and received signals can be reflected as the frequency offset (red line). Since we can know the frequency slope and sound speed in advance, we can determine the corresponding time delay and distance of the object.

Theoretically, if we send the upchirp whose frequency initiates at f_0 and linearly ascends to f_0+B/T, the transmitted signal, $S_T(t)$, can be mathematically represented by

$$S_T(t) = \cos\left(2\pi(f_0 t + \frac{B}{2T}t^2)\right), \tag{1}$$

where f_0 is the base frequency, B indicates the bandwidth, and T is the period of the chirp. Subsequently, the received signal, $S_R(t)$, maintains the same phase but with a variation in amplitude, α, and a time shift, τ:

$$S_R(t) = A\cos\left(2\pi\left(f_0(t-\tau) + \frac{B}{2T}(t-\tau)^2\right)\right). \tag{2}$$

When we apply the product-to-sum identity to the $S_T(t)\cdot S_R(t)$ (*i.e.*, $\cos A\cos B = 0.5(\cos(A-B)+\cos(A+B))$), and utilize

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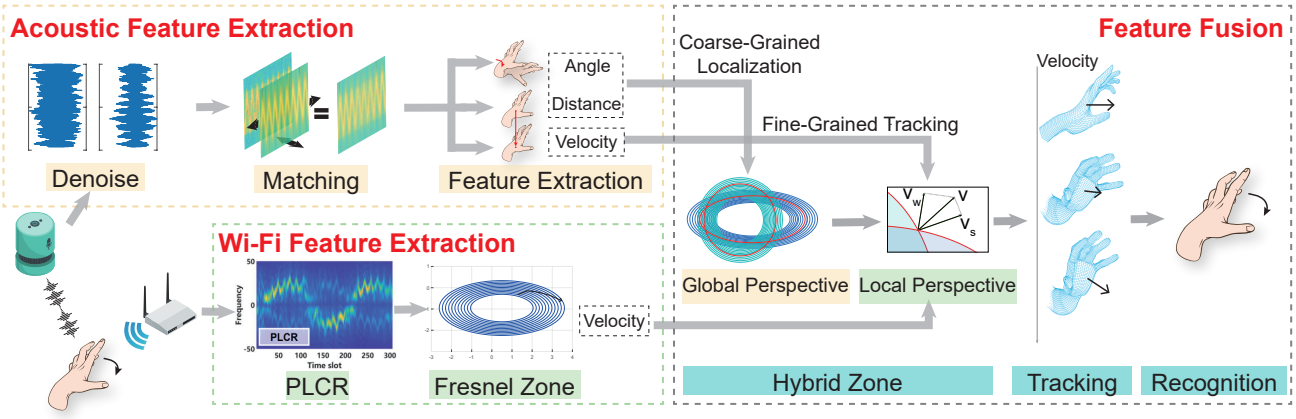


Fig. 4: System overview.

a low-pass filter to filter out the high-frequency signal, we have

$$S_L(t) = \frac{1}{2} A \cos\left(2\pi \frac{B\tau}{T} t + 2\pi\left(f_0\tau - \frac{B\tau^2}{2T}\right)\right). \quad (3)$$

The frequency of the resulting signal $S_L(t)$ is $B\tau/T$, where τ is the propagation time of the acoustic signal. Finally, we can calculate the corresponding propagation distance through $d = \tau \cdot c_s/2$, where c_s is the sound speed.

B. Observations and Insights

Observation 1: Rapid movement of the target can affect the accuracy of distance assessment. The accuracy of distance assessment can be influenced by the rapid movement of the target due to the combined effects of the Doppler effect and the target's distance. The target's distance causes a horizontal shift τ of the signal, resulting in a corresponding frequency shift $f_d = B\tau/T$. Additionally, the Doppler effect causes the signal to have a vertical downward shift by $f_v = 2f \cdot v_s/c_s$, where v_s denotes the velocity of the target. Consequently, the frequency shift should be considered as $f_d + f_v$ in Fig. 5(a), rather than just $f_d = B\tau/T$ as discussed in Eq. (3). To investigate the potential impact of f_v on the measurement of f_d, we examine their ratio, denoted as f_v/f_d, under various scenarios. Fig. 3(b) illustrates this ratio, with the X-axis representing the distance and the Y-axis representing the velocity, while the values denote f_v/f_d. The results demonstrate that a higher ratio of velocity to distance v/d leads to a larger f_v/f_d, indicating that the effect of f_v on f_d is not negligible. Mathematically, we have:

$$\frac{f_v}{f_d} = \frac{2fv_s}{c_s} \cdot \frac{Tc_s}{2Bd} = \frac{fT}{B} \cdot \frac{v_s}{d} = k_{coe} \cdot \frac{v_s}{d}. \quad (4)$$

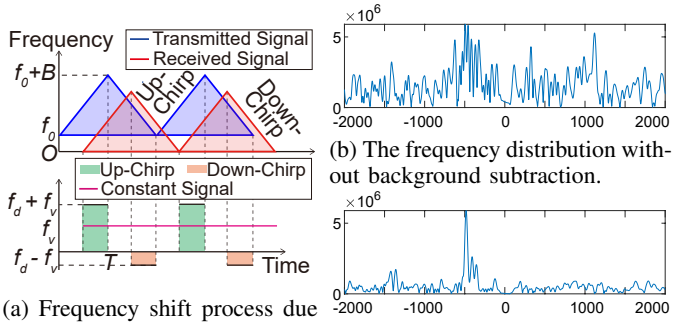
This affirms v_s/d is directly proportional to f_v/f_d, where $k_{coe} = fT/B$. Extracting both f_d and f_v from a single defining equation in the signal poses a significant challenge due to their interdependence.

Insight 1. One straightforward solution is to explore the integral relationship between distance and velocity at different time slots. However, current methods [7] still encounter the problem of ambiguity, where a unique solution cannot always be obtained. To address this issue fundamentally, we draw inspiration from the scaling factor in Eq. (4). Since the speed and distance of the person do not change significantly in a short period (*e.g.*, less than 0.1 s), the ratio v_s/d remains relatively constant. Therefore, adjusting the scale factor $k_{coe} = fT/B$ is equivalent to modifying the ratio of f_v to f_d. By designing chirps with respect to different values of f, T, and B, we can mix f_d and f_v in various proportions. This approach lays the foundation for accurately separating these two variables and resolving the challenges associated with their extraction.

Observation 2: Applicable scenarios of pure acoustic or Wi-Fi-based sensing methods are limited. i) Acoustic sensing requires that the user faces the smart speaker at a narrow angle, which significantly limits the applicable scenarios. The essence of acoustic Doppler velocity is the projection of the target's true velocity onto the line connecting the target and the acoustic equipment. When the target is not directly facing the acoustic device, the value of this projection is weakened. In some cases, we may not be able to detect the Doppler velocity when the target moves along the tangent of the circle. ii) Wi-Fi sensing requires known location information to extract location-independent features. This is because we can observe varying features at different locations even with respect to the same behavior. The underlying mechanism behind this phenomenon is that the distribution of Fresnel zones at different locations is varying in Fig. 2(a), where a denser Fresnel zone results in more pronounced signal changes.

Insight 2. Leveraging the distinct attributes of Wi-Fi and acoustic signals, we developed the "Hybrid Zone", combining a circular acoustic signal model with an elliptical Wi-Fi-based Fresnel zone model, which consists of two stages: i) In the *global hybrid zone*, the acoustic signal pinpoints the target's absolute position, directing our focus to the specific Fresnel ellipse and acoustic circle for signal distribution. ii) The *local hybrid zone* uses Wi-Fi and acoustic speeds, which are projections of actual speed, and applies a theoretical method for detailed speed synthesis. This dual-stage design mitigates the limitations of acoustic signals, like narrow angles and proximity needs. Implementing the hybrid zone enhances the accuracy and reliability of velocity, distance, and angle measurements, and improves object tracking in various scenarios.

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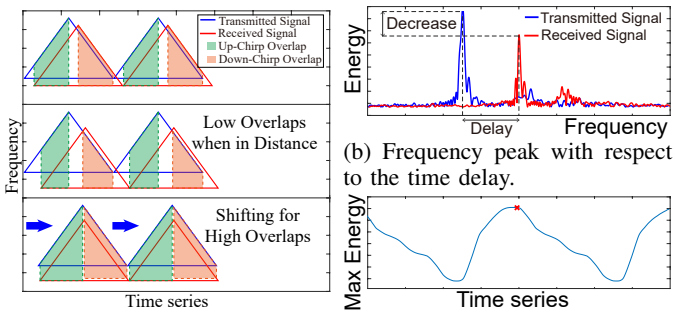


(a) Frequency shift process due to distance and velocity effects during up and down-chirps.

(b) The frequency distribution without background subtraction.

(c) The frequency distribution With background subtraction.

Fig. 5: Background denoise process.



(a) Change of effective overlap with distance.

(b) Frequency peak with respect to the time delay.

(c) Find the time delay with maximum energy.

Fig. 6: Signal enhancement process.

IV. System Designs

A. System Overview

The system illustrated in Fig. 4 is composed of three main components: acoustic feature extraction, Wi-Fi feature extraction, and feature fusion.

1) Acoustic Feature Extraction: To effectively sense rapid motions over long distances, our system employs a triangular chirp signal. This signal is specifically designed to differentiate the impacts of speed and distance on the acoustic signal. We've developed a comprehensive theoretical model that enhances sensing accuracy by leveraging the chirp signal's characteristics in both frequency and time domains. This model enables precise manipulation of the signal for more accurate analysis.

Additionally, our system utilizes a sophisticated microphone array. This array is crucial for determining the angle of the target, which is then used to estimate the target's approximate global position. We apply this coarse-grained positional data in our feature fusion module, setting the stage for advanced velocity synthesis and detailed object tracking.

2) Wi-Fi Feature Extraction: Our methodology for extracting Wi-Fi features commences with calculating the Channel State Information (CSI) ratio. This step is crucial for removing phase offset and highlighting the Wi-Fi signal's dynamic aspects. Subsequently, we apply the Short-Time Fourier Transform (STFT) to detect Doppler frequency shifts. These shifts are converted into the Path Length Change Rate (PLCR), providing a direct measure of the moving object's dynamic trajectory.

Additionally, we incorporate the Fresnel zone concept to refine our analysis further. This involves analyzing how the Wi-Fi signal interacts with multiple ellipsoids, a strategy that significantly improves our capability to discern and interpret signal variations related to object movements.

3) Feature Fusion: In the final stage of our system, we integrate the data from both acoustic and Wi-Fi feature extractions. This integration, which we term 'hybrid zone', splits the process into two perspectives: global and local.

From a global perspective, we use the angle and distance information acquired from the acoustic system to gauge the object's approximate position. On the local scale, we merge the velocity components identified by both the Wi-Fi and acoustic signals. This combined data allows us to create a synthesized, accurate measure of the user's actual speed, crucial for fine-grained tracking.

Finally, we fuse these global and local features in a time series format. This fusion is the backbone for our custom-designed classifier, which is capable of recognizing and distinguishing between different gestures. The classifier serves as the culmination of our system, turning complex data into actionable insights for gesture recognition.

B. Acoustic Feature Extraction

In this stage, we design a series of methods to realize acoustic feature extraction, and answer the following three problems: i) how to remove the interference of background noise? ii) how to realize long-distance sensing? iii) how to accurately determine the distance and speed? Next, we begin with our designed acoustic signals and then illustrate how to address the above-mention problems.

1) Sensing signals transmission: According to **Insight 1**, to distinguish the frequency offset caused by distance and speed, the crux is to design chirps with respect to different f, T, and B. In this paper, we utilize the triangular acoustic signals in Fig. 5(a), which consists of up-chirps, down-chirps, and single-frequency signals. To illustrate why our proposed chirp is valid, we take Fig. 5(a) as an example. Specifically, the effect of distance and speed on the frequency difference is caused by translation in the horizontal and vertical directions, respectively. Intuitively, these two translations have different effects on the up-chirp, down-chirp, and single-frequency signals. For the up-chirp, the resulting frequency shift is $f_d + f_v$; For the down-chirp, the resulting frequency shift is $f_d - f_v$; For the single-frequency signals, the resulting frequency shift is f_v. Hence, we can distinguish f_d and f_v.

Mathematically, the transmitter signal can be represented by

$$s_t(t) = \begin{cases} cos(2\pi f_0 t) + \cos\left(2\pi(f_0 t + \frac{Bt^2}{2T})\right), t \in \mathbb{S}_1 \\ cos(2\pi f_0 t) + \cos\left(2\pi((f_0 + B)t - \frac{Bt^2}{2T})\right), t \in \mathbb{S}_2 \end{cases} \tag{5}$$

where $\mathbb{S}_1 = [2nT, 2nT + T]$ and $\mathbb{S}_2 = [2nT + T, 2(n + 1)T]$ respectively represent the first half and the second half of the periodic signal, and n denotes the n-th periodic signals. Next, we propose our method to extract acoustic features.

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2) Background denoise: Aiming to address the first problem, we design the background denoise method. The fundamental principle of acoustic sensing is to analyze the feature of the signal reflected by the target. In this case, the direct signal, as well as the signal reflected by other objects, can act as interference and hinder our analysis. As illustrated in Fig. 5(b), multiple frequency peaks are respectively corresponding to the direct and dynamic signals. It is evident that the power of the direct signal is stronger compared to the reflected signal, as the latter undergoes greater energy attenuation. To mitigate the effects of these path deviations, we establish a ground truth by collecting acoustic signals with the object at a fixed, stationary distance. In particular, the process starts by handling each chirp size, which entails removing the background signal from the initial data. This approach effectively isolates the signal associated with the reflection path, leaving us with data solely from the target. After we perform the background denoise, the resulting spectrogram is shown in Fig. 5(c), where the direct signal is significantly suppressed, and we can obtain accurate frequency shifting with respect to the dynamic signal.

3) Signal enhancement: To address the second problem, we design a signal enhancement method. In practice, acoustic signals are attenuated as they travel through the air, making it challenging to obtain accurate features over long distances. This problem becomes more serious for our hybrid chirp signals. Specifically, we derive location-related features by analyzing the frequency shift between the transmitted and received signals. However, as the distance between the smart speaker and the object increases, the received signal experiences a longer delay, leading to a reduction in the effective overlap between the transmitted and received signals, as illustrated in Fig. 6(a). Accordingly, the frequency peak with respect to the object will decrease in Fig. 6(b). In order to enhance the perceived signal at long distances, our basic idea is to actively add delay to the source signal, so that we can obtain a higher overlap area, as illustrated in Fig. 6(a). To find the optimal overlap area, we can iterate over the different time delays t_d and record the corresponding peak value $P(t_d)$. Fig. 6(c) illustrates the relationship between t_d and $P(t_d)$, where we can observe significant differences for different delays. Intuitively, a larger effective overlap area leads to higher $P(t_d)$ and more accurate features, and we have $\hat{t}_d = \arg\max_{t_d} P(t_d)$. In addition, when we involve such a delay, it is equivalent to translating the signal horizontally. Since f_d is also reflected as a translation of the signal in the horizontal direction, its estimation will be affected. To recover the feature, we record such extra delay $\hat{t}_d$ and then compensate the corresponding offset to the resulting distance, which is calculated by $\hat{t}_d c_s$.

4) Feature extraction: After enhancing the reflected signal, we analyze the time-frequency translation of the triangular signal to resolve location-related features. As depicted in Fig. 5(a), the equations for the periodic signals are given by:

$$\begin{cases} f_{up}(n) = f_d(n) + f_v(n), \\ f_{down}(n) = f_d(n) - f_v(n), \\ f_{single}(n) = f_v(n), \end{cases} \tag{6}$$

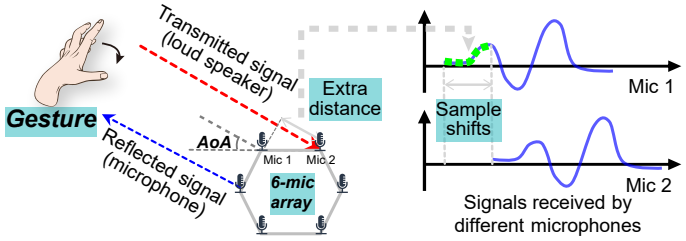


Fig. 7: The principle of AoA estimation.

where n denotes the n-th periodic signal, and $f_d(n)$ and $f_v(n)$ are defined as

$$\begin{cases} f_d(n) = d(n)B/(Tc_s), \\ f_v(n) = v_s(n)f/c_s. \end{cases} \tag{7}$$

To determine $d(n)$ (distance) and $v_s(n)$ (velocity), we analyze $f_{up}(n)$, $f_{down}(n)$, and $f_{single}(n)$. In the time domain, we consider the integral relationship $d(n) = d(n-1)+2v_s(n-1)T$ to further refine our calculations.

For acoustic velocity estimation, we combine time-frequency domain data using this equation

$$v = \sqrt{v1 \cdot v3} + \min(e^{\alpha|v_2 - \sqrt{v1 \cdot v3}|}, 1)(v_2 - \sqrt{v1 \cdot v3}), \tag{8}$$

where α is a positive constant, and the components v_1, v_2, and v_3 are defined as

$$\begin{cases} v_1 = (f_{up} - f_{down})c_s/(2f), \\ v_2 = \dfrac{\mathrm{d}}{\mathrm{d}t}\left(\dfrac{(f_{up} - f_{down}) \cdot c_s T}{2B}\right), \\ v_3 = f_{single}c_s/(2f). \end{cases} \tag{9}$$

Equation (8) integrates these three velocity components, considering factors such as frequency shifts and time differentials, to provide a comprehensive and accurate estimation of the target's velocity.

5) AoA estimation: Currently, it is insufficient to obtain the global location of the target with only knowledge of the distance. Hence, we here propose our method to calculate the angle of arrival (AoA) of the target. The basic principle of AoA calculation is similar to [7], where we can utilize the microphone array and the MUSIC [4], [27] to determine the AoA of the target. As illustrated in Fig. 7, since the target distance is much larger than the distance between the microphones, the signal arriving at the microphone array can be viewed as a series of parallel lines. Assuming this premise, it is possible to detect a time delay in the signals captured by various microphones when the signals reach them from different angles. In our proposed system, the microphone array can capture both direct and reflected signals, where the direct signal is stronger than the reflected signal. Hence, we can not detect the time delay of the raw signals directly. Instead, we can first apply our proposed background denoise method to remove the direct signal, and then apply the proposed method in Fig. 7 to detect the sample shifts and the AoA. In practice, the AoA is not so fine-grained to depict the user's gesture, hence, we will focus on the local hybrid zone later.

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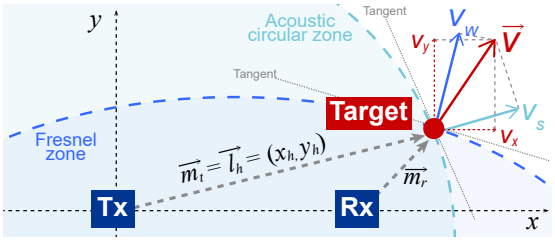


Fig. 8: Feature fusion.

C. Wi-Fi Feature Extraction

In this phase, we extract Wi-Fi features from raw Channel CSI readings through physical processing. The CSI signal encapsulates various path elements, including static Line-of-Sight (LoS), environmental reflections, and dynamic paths from objects. The CSI at frequency f and time t is expressed as:

$$H(f,t) = H_s(f,t) + A(f,t)e^{-j2\pi\frac{d(t)}{\lambda}},$$

where $H_s(f,t)$ is the static component, and $A(f,t)$, λ, and $d(t)$ represent the attenuation, wavelength, and dynamic path length, respectively. Movement changes the path length of reflected signals, inducing a DFS. To determine DFS, we first compute the CSI ratio, then apply the STFT. We identify the DFS by extracting the highest energy frequency at each moment from the STFT spectrum, denoted as $f_{doppler}(t)$. The Fresnel model indicates that a full cycle in CSI magnitude equates to one wavelength change in path length. Hence, the PLCR of reflected signals is $v_w(t) = f_{doppler}(t)\lambda$. However, v_w doesn't exactly represent human velocity. To correlate PLCR with actual speed v, we consider v_w as the projection of v on the Fresnel zone's normal line. With pre-known transmitter ($\vec{l_t} = (x_t, y_t)^T$) and receiver ($\vec{l_r} = (x_r, y_r)^T$) positions, and the target's current position ($\vec{l_h} = (x_h, y_h)^T$) and velocity ($\vec{v} = (v_x, v_y)^T$), we can define PLCR v_w.

D. Feature Fusion

The purpose of this stage is to fuse the acoustic and Wi-Fi features based on hybrid zone. In particular, the acoustic feature extraction stage provides $\{\theta, d, v_s\}$, while the Wi-Fi feature extraction stage provides $\{v_w\}$. To fuse the feature, we first use the interpolation method to synchronize acoustic and Wi-Fi features, and then propose how to use the hybrid area to fuse multi-modal features. Finally, we draw the trajectory of the target and realize gesture recognition.

1) Synchronization: To effectively analyze the current state using both Wi-Fi and acoustic signals, we need to ensure two crucial elements. First, the data from both signals must be synchronized, meaning that they should start and end concurrently. Second, since these two types of signals are recorded at different rates, it's necessary to manipulate the data so that they share the same time intervals. To accomplish this, we standardized both the Wi-Fi and the acoustic signals at a common sampling rate of 100 samples per second. The interpolation was then used to achieve uniform values. This process resulted in synchronized data with matching time intervals, thereby for further processing and analysis.

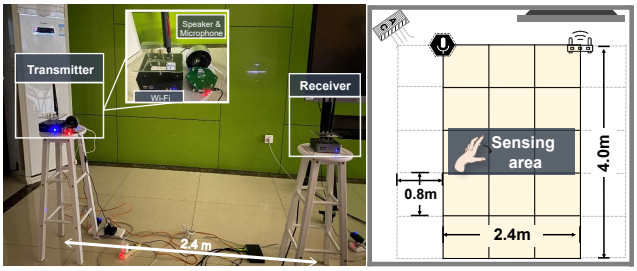


Fig. 9: Experimental scenarios.

2) Hybrid Zone: In the global hybrid zone, we use the position of the voice device as the origin to establish a coordinate system. If the current target position is $\vec{l_h} = (x_h, y_h)^T$, we have $x_h = d\cos\theta$ and $y_h = d\sin\theta$ As shown in Fig. 8, when we focus on the local hybrid zone based on the above location, supposing that the target velocity is $\vec{v} = (v_x, v_y)^T$, the velocity detected by the acoustic system is

$$v_s = \alpha_x v_x + \alpha_y v_y, \tag{10}$$

where

$$\alpha_x = \frac{x_h}{\|\vec{l_h}\|},\ \alpha_y = \frac{y_h}{\|\vec{l_h}\|}. \tag{11}$$

Recall that we have $v_s = \alpha_x v_x + \alpha_y v_y$ and $v_w = \beta_x v_x + \beta_y v_y$. Then we can obtain the following equation:

$$\vec{v}_{obs} = \mathcal{L} \cdot \vec{v}, \tag{12}$$

where $\vec{v}_{obs} = [v_s, v_w]^T$, and $\mathcal{L} = \begin{bmatrix} \alpha_x & \alpha_y \\ \beta_x & \beta_y \end{bmatrix}$. In order to derive the user's velocity, we can solve $\vec{v}$ in Eq. (12) by

$$\vec{v} = (\mathcal{L}^T\mathcal{L})^{-1}\mathcal{L}^T\vec{v}_{obs}. \tag{13}$$

This allows us to calculate the target's true velocity $\vec{v}$ from the observed values, and forms the foundation for deriving accurate velocity estimations through hybrid zone.

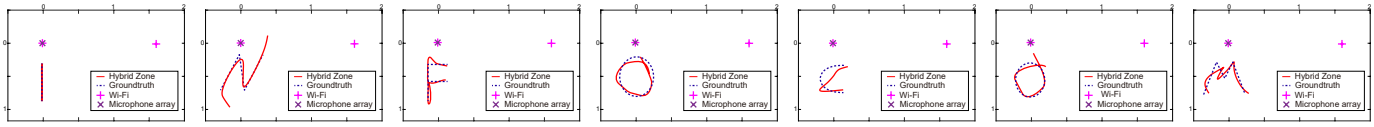
V. Evaluation

This section delineates our experimental settings, spanning hardware and software considerations, and subsequently discusses the performance assessment of our system under varying conditions.

A. Implementation

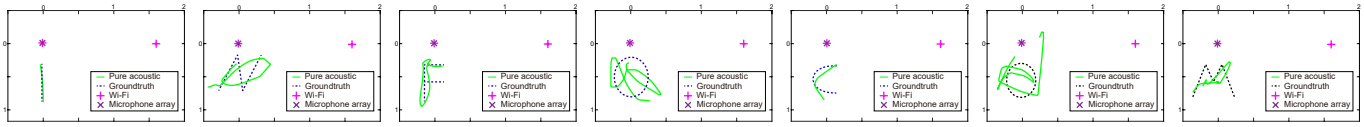
Our hardware setup, as shown in Fig. 9, encompasses various components necessary for our experiments. The Wi-Fi component of our system uses two devices equipped with Intel 5300 NICs: a single-antenna transmitter and a three-antenna receiver. For acoustic sensing, we employ a Seeed Respeaker circular array with six microphones, comparable to commercial smart speakers like Google Home and Amazon Echo. The microphone array, designed in a regular hexagon shape with a side length of 4.75cm and a sampling rate of 48kHz, is co-located with the Wi-Fi transmitter to support both acoustic and Wi-Fi functionalities. Both the microphone array and Wi-Fi transmitter are connected to a laptop for data collection, with the receiver positioned at a distance to enable effective sensing. For software, we explain the tools and environments used for data collection and analysis.

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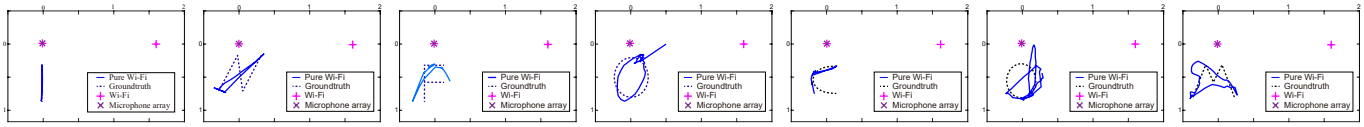
(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 10: Tracking examples of different trajectories with Hybrid Zone.



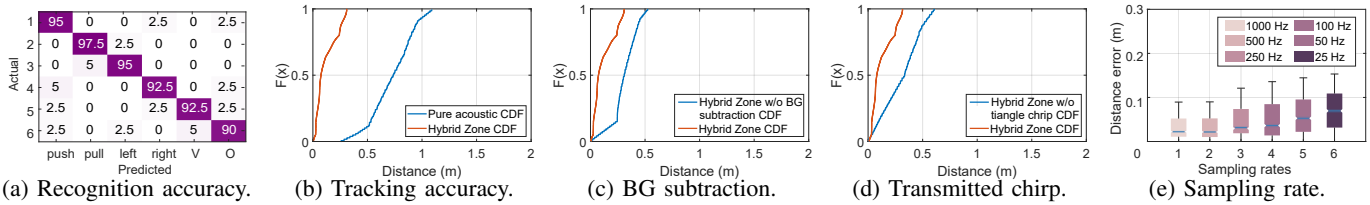
(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 11: Tracking examples of different trajectories with pure acoustic methods (SpeakerGesture [9]).



(a) Gesture 'I'. (b) Gesture 'N'. (c) Gesture 'F'. (d) Gesture 'O'. (e) Gesture 'C'. (f) Gesture 'O'. (g) Gesture 'M'.

Fig. 12: Tracking examples of different trajectories with pure Wi-Fi method (Widar2.0 [28]).



(a) Recognition accuracy. (b) Tracking accuracy. (c) BG subtraction. (d) Transmitted chirp. (e) Sampling rate.

Fig. 13: Overall performance.

B. Overall System Performance

Assessing the Entire System Performance. To evaluate our proposed method's efficacy, we conducted an experiment where the term "INFOCOM" was sketched. Each letter of "INFOCOM" is represented in a series of figures (Fig. 10, 11, 12), with each figure illustrating a different letter. The gestures for 'I' to 'M' are showcased in three scenarios: with and without implementing the hybrid zone. The results are compared to a method relying solely on acoustic and Wi-Fi sensing, emphasizing the hybrid zone's effectiveness in overcoming the limitations of each individual technique.

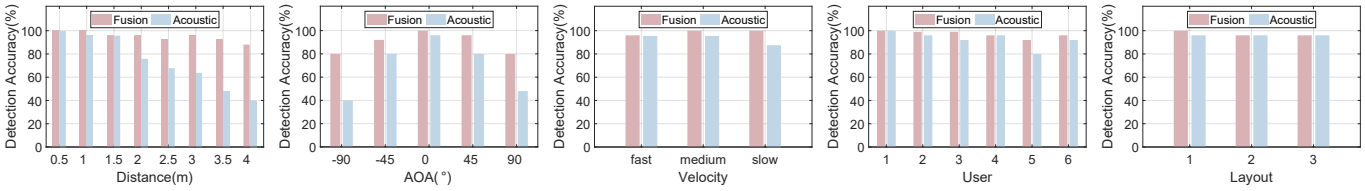
Evaluation of Gesture Recognition Performance Beyond mere letters, we have identified six additional gestures for a more comprehensive evaluation: 'push', 'pull', 'leftward', 'rightward', O-shape, and V-shape. For each of these gestures, we collect 40 data samples. Our objective is to leverage hybrid zone to understand the unique signal patterns of these gestures. To better demonstrate hybrid zone's performance, we present the results in two aspects: classification and tracking. The results of the recognition process are showcased in Fig.13(a) and Fig.13(b), representing the classification result and tracking accuracy, respectively. We aim to accomplish the classification by training a machine-learning model for recognition. As we have already established a theoretical model in this paper, we can generate some simulated data to train the neural network for gesture recognition. In Fig.13(a), we employ a confusion matrix to present the overall result of classification. We observe that the 'pull' gesture achieves the highest accuracy of 97.5%, and the O-shape gesture has the lowest accuracy of 90%. All of the six gestures maintain an accuracy above 90%. In Fig.13(b), we use a Cumulative Distribution Function (CDF) figure to show the tracking accuracy. We contrast this with the pure acoustic sensing method, where hybrid zone attains 90% accuracy within an error of 0.26m, while the acoustic method has an error of 0.8m.

C. Analysis of Variable Impact on System Performance

Impact of Background Subtraction. We examine background subtraction in enhancing our system's performance. As discussed earlier, most existing acoustic methods estimate the target position directly from the obtained frequency shift, neglecting any frequency shifts that aren't induced by distance. To address this, we incorporate a repetitive check after the coarse-grain distance estimation in our method. The results are shown in Fig. 13(c). The blue and orange lines follow similar trends, with the former appearing as a horizontal shift of the orange line. This supports our approach's effectiveness in reducing errors caused by incorrect distance estimations.

Impact of Chirps. To investigate how different chirp configurations affect our system's performance, we considered the use of triangle chirps. The reason we incorporate a down-chirp and a constant frequency signal, as mentioned earlier, is to better separate the factors of velocity and distance. Given that most existing acoustic-based research employs a linear up-chirp as the transmitted signal, we use the up-chirp portion

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(a) Impact of distance. (b) Impact of angle. (c) Impact of speed. (d) Impact of user. (e) Impact of layout.

Fig. 14: Impact of diverse parameters.

of the triangle chirp for comparison. In other words, We focus solely on the first half of the triangle chirp, specifically the up-chirp, as depicted in Fig. 13(d), while omitting the second half. In this figure, the two lines closely follow each other, with the blue line slightly to the right of the orange one.

Impact of Wi-Fi Sampling Rate. We assess how different Wi-Fi sampling rates, ranging from 20Hz to 1000Hz, impact our system's performance. This evaluation is vital for gauging the system's adaptability in diverse Wi-Fi conditions. The outcomes of these assessments are presented in Fig. 13(e). The findings from these tests provide insights into the optimal sampling rates for achieving the best gesture recognition accuracy under various environmental conditions.

Impact of Distance. We conduct tests to investigate the variation in system performance with changes in the distance between the sensor and the subject. We consider eight different distances from 0.5 meters to 4 meters. For each distance, we ask a volunteer to repeat a gesture five times, with five repetitions per trial. As shown in Fig. 14(a), hybrid zone (blue bars) maintains an accuracy of 90% up to 3.5 meters. In contrast, pure acoustic sensing (red bars) fails to maintain its sensing capabilities (falling below 90% accuracy) after 1.5 meters. These results highlight the superiority of hybrid zone in capturing gestures over long distances.

Impact of Angle. We also test the effectiveness of hybrid zone under varying angles from -90 degrees to 90 degrees. Similar to the distance test, we ask the volunteer to repeat gestures 25 times at each angle. Unlike the acoustic-based methods that only achieve acceptable performance when facing the target directly, hybrid zone takes advantage of its velocity fusion capabilities. This allows it to yield more precise distance and direction estimates and thus more accurate gesture predictions. The results in Fig. 14(b) show that hybrid zone maintains an accuracy higher than 80% at all angles.

Impact of Gesture Execution Speed. We examine the effect of the speed at which a gesture is executed on system performance. We ask the volunteer to perform the same gesture at three different speeds: slow, medium, and fast. As shown in Fig. 14(c), hybrid zone consistently maintained an accuracy above 95% across all speeds. However, purely acoustic methods underperformed at slow speeds. This drop in performance can be attributed to the slower speed, resulting in negligible differences between consecutive timestamps. This makes it difficult for the system to discern directional changes using pure acoustic sensing alone.

Impact of User Variability. We conduct tests to measure the effect of user variability on our system's performance. We enlisted six volunteers to perform gestures 25 times each. The results, presented in Fig. 14(d), show how the performance of the system varied among users. All users achieved an accuracy higher than 90%, with User 1 having the highest accuracy at 100%, and User 5 having the lowest accuracy at 90%. Although acoustic signals also demonstrate good robustness across various users, the results may not be as reliable when users perform gestures differently, with an accuracy falling below 80% as shown in the figure.

Impact of Device Placement. We evaluate the implications of the placement of the sensor device on our system's effectiveness. Our goal is to demonstrate that hybrid zone can maintain its robustness even in varying environments. The experiment is conducted in three different layouts, with environmental factors such as air conditioners and televisions, as shown in Fig. 9. In the three layouts, the distances between the Wi-Fi transmitter and receiver are respectively 0.8, 1.6, and 2.5 meters. The result is shown in Fig. 14(e). Hybrid zone maintains an accuracy above 90% across all three layouts. Interestingly, we observed that the pure acoustic method seems not to be influenced by different layouts. This is because the layout only changes the distance between the Wi-Fi routers, while the acoustic system remains unaffected. In other words, changes in the layout generally do not affect the placement of the acoustic techniques. Therefore, the sensing accuracy of the acoustic methods remains largely unchanged.

VI. Conclusion

This paper presented "Hybrid Zone," a novel system combining acoustic and Wi-Fi signals for gesture recognition. With a recognition accuracy of 93.75%, our system operates effectively using a single acoustic setup and a Wi-Fi transmitter-receiver pair, distinguishing itself by not requiring prior knowledge of the target's initial position. Extensive experiments have emphasized both the accuracy and effectiveness of Hybrid Zone, while also showcasing its distinct capacity to combine the benefits of acoustic and Wi-Fi technologies. A key aspect of our system is its adaptability in diverse real-world applications. Hybrid Zone proves flexible in environments ranging from smart homes, which demand low intrusiveness and high precision, to industrial contexts where resistance to environmental noise is essential. In conclusion, Hybrid Zone stands out for its high accuracy in gesture recognition, and versatility across different environments, marking it as a pivotal advancement in the field.

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